Gregory R. Warnes, Ph.D.

SENIOR R PACKAGE DEVELOPER

CONTACT

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EDUCATION

Ph.D., Biostatistics
University of Washington
2000

M.Sc., Biostatistics
University of Washington
1997

B.Sc., Statistics
Brigham Young University
1995

A.Sc., Computer Science
Weber State University
1989

AWARDS

Pfizer Global Achievement Award
2004

Eagle Scout
1987

LANGUAGES

- English (Native)
- French (Professional Working)

SUMMARY

Senior Bioinformatics Software Engineer with over 25 years of experience specializing in high-performance R Package Development and robust Software Architecture for complex Genomics research. Currently provides expert consulting on experimental design and RNA-Seq analysis while maintaining a portfolio of enterprise-grade R tools and Bioconductor-aligned workflows. Previously created the MiDAS Genomic Workflow System at Pfizer, automating RNA microarray analysis from raw data QC to final biological interpretation, and led the conversion of legacy clinical trial data into CDISC standards. Reduces Technical Debt across large-scale projects by implementing rigorous CI/CD practices and the S4 object system to ensure codebase scalability. Seeking to leverage deep expertise in R package development and statistical computing to advance Genentech's bioinformatics ecosystem and accelerate biological discovery.

SELECTED ACHIEVEMENTS

- Co-founded Revolution Analytics (acquired by Microsoft), creating the industry standard for enterprise-grade R deployment. Architected RStatServer to operationalize statistical models.
- Architected the MiDAS Genomic Workflow System at Pfizer, an automated pipeline for microarray conception, QC, normalization, and analysis. Earned the 2004 Pfizer Global Achievement Award.
- Author of 20+ R packages (including genetics, qvalue, gplots, gmodels) with over 7,200 citations.
- Built a cloud-based genomic workflow tool for Medidata's Rave platform, enabling seamless management and modeling of genomic clinical trial data.

TECHNICAL SKILLS

Core Expertise

- Algorithmic Architecture (Expert)
- Analytic Software Development (Expert)

General

- Bioconductor
- Cloud Object Storage
- GitLab
- HDF5
- JWT
- OAuth2
- REST API
- Rcpp
- S4 Classes
- System Refactoring
- Technical Debt Management

Infrastructure & Cloud Platforms

- Data Architecture (Expert)
- Continuous Integration and Continuous Deployment (Advanced) ≡ CI/CD
- Amazon Web Services (Advanced) ≡ AWS
- Docker (Advanced)
- Google Cloud (Advanced)

EXPERIENCE

Owner & Chief Scientist — **Warnes Innovations, LLC**

2013-01 – present

- Statistical Genomics: Provided expert consulting on experimental design and statistical analysis for multiple RNA-Seq projects.
- Supported functional genomics workflows and interpretation of biological patterns.
- Broadband Analyzer: Designed and deployed a scalable data pipeline and full-stack architecture for large-scale geospatial data analysis.
- Bank Advertising ML Model: Developed and validated statistical models to project consumer behavior and optimize marketing spend.

Principal Data Scientist — **Torqata Data and Analytics, LLC**

2018-08 – 2020-06

- ML Model Productionization: Invented and implemented a predictive methodology combining spatial demographics and sales volume.
- Integrated Machine Learning models into production SaaS platform (Category Compass).
- Translated complex customer inventory challenges into specific analytic features and workflows.

Principal Data Scientist — **Medidata Solutions Inc**

2016-01 – 2018-12

- Bioinformatics Pipelines: Developed a scientific workflow system for genomic clinical trial data, covering QC, normalization, and integration within the Rave digital health platform.
- Data Standardization & QC: Led conversion of legacy trial data into CDISC standards using Natural Language Processing (NLP), increasing data ingestion success by 12x.
- Executive Analytics: Provided statistical support and custom ad-hoc analysis for corporate leadership, delivering data-driven insights to inform strategic decision-making.
- Statistical Consulting & Mentoring: Provided statistical consulting and mentoring to the data science division, developing team capabilities and ensuring statistical rigor across projects.

Programming & ML Frameworks

- Bash (Expert)
- C/C++ (Advanced), Python (Expert)
- R (Expert)

Scientific & Bioinformatics

- Genomics (NGS, GWAS) (Expert), RNA-Seq, scRNA-Seq (Expert), Multi-omics Integration (Expert)
- Pipeline Development (Expert)

Senior Principal Biostatistician — Boehringer Ingelheim Pharmaceuticals Inc

2014-01 – 2016-12

- Multi-omics Analysis: Provided statistical leadership for Non-clinical and Safety divisions, analyzing high-resolution and single-cell genomic data (scRNA-Seq).
- Scientific Interpretation: Reviewed experimental data to identify gaps in evidence, ensuring rigorous statistical validity for target discovery and safety testing.
- Phase Gate Review: Reviewed Phase Gate submissions for data quality and statistical rigor, ensuring compliance with regulatory standards and scientific best practices across drug development programs.

Senior Expert Modeler — Novartis Institutes for BioMedical Research

2011-01 – 2013-12

- Interactive Clinical Trial Optimization Tool: Developed an interactive and statistically rigorous web-based tool for assigning clinical study patients and sites to countries, optimizing for time to completion and cost while accounting for country, therapeutic area, indication, study phase, and study length effects on costs, delays, and recruitment rates under resource and other constraints.
- PK/PD Clinical Trial Simulation: Conducted clinical trial simulation using PK, PD, patient distribution, and trial feature information primarily for ophthalmology studies, enabling evidence-based trial design and protocol optimization.

Assistant Professor of Biostatistics — University of Rochester

2006-05 – 2011-12

- Led software development at the Center for Biodefense Immune Modeling (DEDiscover).
- Taught graduate courses on Statistical Computing and Bayesian Modeling.

Co-founder Revolution Analytics

2005-06 – 2006-12

- Conceived and drove the commercial R initiative at Scientific Computing Associates, bringing the idea of commercially-supported R to the leadership team.
- Created product specifications and marketing materials for enterprise-grade R deployment solutions.
- Coined the name 'REvolution Computing' (R + Evolution) to evoke both the R programming environment and technological progress.
- Advised on specifications and code development for NetWorkSpaces and Sleigh frameworks for R and Python distributed computing.
- Company ultimately acquired by Microsoft, validating the commercial viability of enterprise R solutions.

Associate Director of Non-clinical Statistics Pfizer

2000-10 – 2006-04

- Created the MiDAS Genomic Workflow System, automating the analysis of RNA microarrays from raw data QC to final biological interpretation.

SELECTED PUBLICATIONS (15)

1. Warnes, Gregory R. et al. (2007) GeneticsDesign: Classes and Methods for Genetics Studies.
2. Warnes, Gregory R. et al. (2005) GeneticsBase: Base Functions for Genetics.
3. Warnes, Gregory R. (2002) RSOAP: SOAP Server for R.
4. Warnes, Gregory R. and Leisch, F. (2002) genetics: Population Genetics Analysis.
5. Warnes, Gregory R. (2005) gtools: Various R Programming Tools.
6. Warnes, Gregory R. (2005) gdata: Various R Programming Tools for Data Manipulation.
7. Warnes, Gregory R. (2005) gplots: Various R Programming Tools for Plotting Data.
8. Warnes, Gregory R. (2005) gmodels: Various R Programming Tools for Model Fitting.
9. Warnes, Gregory R. (2003) RSessionDA: Zope Database Adapter for R.
10. Warnes, Gregory R. (2007) SASxport: Read and Write SAS XPORT Files.
11. Warnes, Gregory R. and Li, F. (2004) ssize: Sample Size Calculation for Microarray Experiments.
12. Warnes, Gregory R. (2003) The Genetics Package. R News.
13. Warnes, Gregory R. (2002) session: R Session Management Tools.

14. Moriera, W. and Warnes, Gregory R. (2004) rpy: Python Interface to R.

15. Warnes, Gregory R. et al. (2001) gregmisc: Greg's Miscellaneous Functions.